

ITCS 3166
Assignment 2

Note: Deadline for this assignment is Sep. 27 by 11:59 PM

1, The following character encoding is used in a data link protocol: A: 01000111 B: 11100011 FLAG: 01111110 ESC: 11100000 Show the bit sequence transmitted (in binary) for the four-character frame A B ESC FLAG when each of the following framing methods is used: (a) Byte count. (b) Flag bytes with byte stuffing. (c) Starting and ending flag bytes with bit stuffing.

2, A bit string, 011110111110111110, needs to be transmitted at the data link layer. What is the string actually transmitted after bit stuffing?

3, One of your classmates, Scrooge, has pointed out that it is wasteful to end each frame with a flag byte and then begin the next one with a second flag byte. One flag byte could do the job as well, and a byte saved is a byte earned. Do you agree?

4, What is the maximum overhead in byte-stuffing algorithm?

5, A channel has a bit rate of 4 kbps and a propagation delay of 20 msec. For what range of frame sizes does stop-and-wait give an efficiency of at least 50%?